Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience owning features end to end: design, build, test, deploy, and operate distributed backend systems on AWS using Java, Python, and Kubernetes, with production oncall and incident response experience.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, SQL, Go, JavaScript, TypeScript.

Distributed Systems: Object-Oriented Design, multi-threaded services, microservices, REST APIs, system design, reliability and scaling.

AWS & Infrastructure: AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash.

Backend & Data: PostgreSQL, MySQL, HashiCorp Vault, access control, audit logging, CI/CD.

Developer Tools: Git, GitLab CI/CD, Jenkins, SonarQube, code review, Azure, Google Cloud Platform.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline recording and replaying agent actions to reproduce Android GUI bugs, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity end to end.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Owned infrastructure automation end to end with Python (Boto3) for **Teamcenter X**, replacing legacy Ruby scripts and cutting deployment time **40%** across AWS environments.
- Handled 44 live production environment upgrades over 4 months as de facto oncall, diagnosing and resolving infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework enforcing policy checks pre-cutover, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault, integrating AWS services for access control.
- Served as Quality Coach and SonarQube Specialist, raising automated code coverage **35% to 78%**; mentored **5 software engineers**.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed, auditable answers to eliminate hallucination in high-stakes decisions.
- Designed **17-tool MCP server** exposing policy-enforced tool-calling for PA extraction and cross-payer comparison, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).